Reward Shaping Is Noise Shaping, and a Model of the Accuracy–Diversity Frontier of RLVR

Project report: results, open questions, running experiments

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Summary

1. **Question.** RLVR methods that modify the advantage claim higher pass@k but are compared at equal training steps, which confounds pass@k with progress in pass@1. We compare at matched pass@1.
2. **Diagnosis.** Per problem, every correctness-only advantage rule has $\mathbb{E}[g_x] = c(p_x)\nabla_\vartheta p_x$; rules differ in the per-problem coefficient and in the second moment. Under the tested protocol, NSR and PSR are equivalent on average at matched pass@1 on two model families, and trace-matched zero-mean sign noise added to PSR reproduces NSR's early pass@1 lag.
3. **Methods.** All interventions inside RL tested so far give at most about +1.2 pp of pass@16 at matched pass@1. A supervised warm start on base-model successes and Dr. GRPO with groups of 32 reach pass@1 outside Dr. GRPO's 150-step range, where matched-pass@1 comparison is undefined.
4. **Frontier model.** A per-problem log-odds model initialized from base-model success counts, with one fitted transfer coefficient, reproduces pass@16 with RMSE 0.85–1.90 pp on three pre-registered frozen-model test sets (nine further rules, OLMo, 1,024 tokens); it fails the registered criterion after the warm start. The extrapolation test (450 steps) is running.
5. **Assessment.** Independent review (Codex): about 5/10 for the diagnosis given the 2026 literature; 7/10 and 8/10 are its scenario assessments if the frontier model passes the remaining tests and if a method improves the frontier across scales.

1 Setting, notation and metrics

Models and data. Qwen2.5-Math-1.5B: MATH training problems, 150 RL steps of 16 problems × 8 rollouts, AdamW 10^{-6}, 512-token answers; evaluation on 511 held-out MATH test problems (answer keys audited). OLMo-2-0425-1B-SFT: 300 steps, two seeds, 256 held-out problems, 1,024-token answers.

Success probabilities and pass@k. $p_x = \Pr_\pi(r = 1 \mid x)$ for problem x, and over N problems

$$P_k = \frac{1}{N}\sum_x \left[1 - (1 - p_x)^k\right], \qquad \widehat{P}_k = \frac{1}{N}\sum_x \left[1 - \binom{n_x - C_x}{k} \Big/ \binom{n_x}{k}\right],$$

with C_x correct answers among $n_x = 32$ independent samples ($\widehat{P}_k$ is unbiased for P_k). pass@1 is accuracy; pass@16 is the diversity measure.

Matched-pass@1 comparison. For a checkpoint with pass@1 a, the reference value

$$E_B(a) = \max\{L_{\text{steps}}(a),\, L_T(a)\}$$

is the larger of the linear interpolations in pass@1 along Dr. GRPO's trajectory (base, steps 50, 100, 150) and along its step-150 temperature sweep ($T = 1, 1.15, 1.3$), each within its supported pass@1 range. The gain is $\widehat{P}_{16} - E_B(a)$, averaged over a run's qualifying checkpoints, then over seeds; intervals: 2,000 bootstrap draws over problems with

the reference recomputed per draw (endpoint clamping reported). Pre-registered bar: +2 pp with the 95% interval above 0 in all seeds.

Strata. Problems grouped by the base model's successes in 32 samples at the stated budget: never ($c = 0$), rare ($1 \le c \le 4$), medium ($5 \le c \le 16$).

2 Diagnosis

Theory. For any rule whose advantages depend only on correctness, per problem $\mathbb{E}[g_x] = c(p_x)\,\nabla_\vartheta p_x$ and $\mathbb{E}\|g_x\|^2 = F_1 D_1(p_x) + F_0 D_0(p_x) + \|\nabla p_x\|^2 X(p_x)$, with F_1, F_0 the mean squared score norms of correct and incorrect answers. Cross-problem reweighting can change the aggregate direction of $\sum_x \mathbb{E}[g_x]$ through shared parameters (Barakat et al., 2026), so "one curve" is an empirical claim. Measured F_0/F_1: 2.14 and 2.54 on OLMo; 2.7–3.4 at initialization and about 1.4–1.6 at steps 50–150 on Qwen. At matched pass@1, P_k depends only on the dispersion of $\{p_x\}$ across problems.

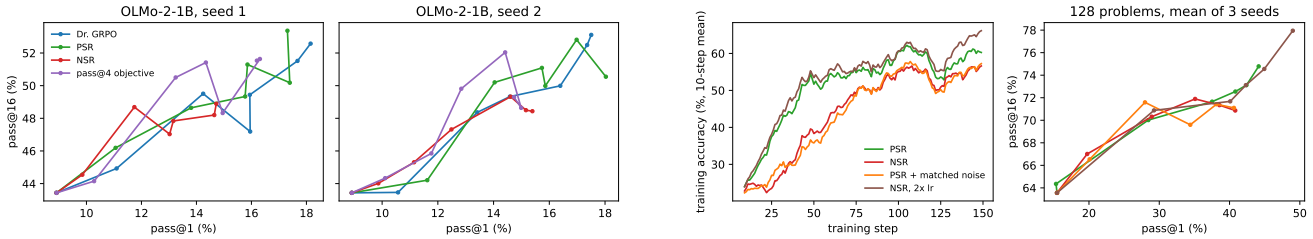


Figure 1: Left: OLMo, five rules in the (pass@1, pass@16) plane. Right: Qwen; training accuracy of PSR, NSR, PSR with trace-matched sign noise, and NSR at twice the learning rate, and their (pass@1, pass@16) trajectories.

Results

1. **Matched-pass@1 equivalence.** On Qwen, PSR, NSR, five constant baselines, the NSR paper's implementation and a pass@4 objective lie within about ±2 pp of one curve. On OLMo, NSR minus PSR at matched pass@1 is −0.30 pp, 90% $[-1.28, +0.77]$: average equivalence within ±2 under the tested protocol (checkpoint-wise equivalence is not established).
2. **Noise injection.** Trace-matched zero-mean sign noise added to PSR reproduced the pooled early pass@1 lag of NSR (speed index $\phi = -0.11$, 90% $[-0.22, -0.05]$, with NSR = 0 and PSR = 1); one seed was heterogeneous, and the scalar noise magnitude is not uniquely identified. NSR at twice the learning rate reaches PSR's original rate; at a matched doubled rate NSR still lags. Under Adam the noise reduces the alignment of steps with the expected direction (13–27%) rather than their size.
3. **Temperature.** Entropy on fixed text changes little (1.081 → 1.064) while entropy on self-generated samples falls 3–4×; a late checkpoint sampled at higher temperature does not exceed the training trajectory (−1.42 pp, 90% $[-2.19, -0.68]$).
4. **Other interventions.** The unlikeliness reward gives +0.14 pp at matched pass@1 (three seeds); entropy bonus, adaptive entropy and entropy floors were tested in a one-seed pilot (no gain; floors and adaptive entropy collapse training).

3 Interventions intended to move the frontier

Intervention (Qwen, three seeds, 511 problems unless noted)	Gain at matched pass@1 (pp of pass@16)	Status
Reward shaping, entropy control, unlikeliness, pass@4 objective	≈ 0	complete
Coverage-weighted replay of own successes (Pre-reg. 12)	$+1.19$, 95% $[+0.54, +1.79]$	complete
Base-seeded replay (Pre-reg. 14)	$+1.02$, 95% $[+0.23, +1.68]$; step-150 pass@1 -5.35	main arms complete
Annealed replay (AR) / MaxRL with group completion (MXGC); pilot, seed 12, 255 validation problems	$+1.30$ / $+1.12$; step-150 pass@1 -6.6 / -6.5 (below the pilot's $+1.5$ gain criterion; outside its -2 pp pass@1 allowance)	interim
Group completion; prefix-guided rollouts (pilot)	–	under evaluation
Supervised warm start on base successes, then Dr. GRPO (S32)	undefined: step-150 pass@1 52.5 vs 44.1	–
Dr. GRPO with groups of 32 (G32)	undefined: step-150 pass@1 52.5 (two seeds)	partial

Numbers marked interim are listed with their computation in `RESULTS_INTERIM_2026-10-23.md`.

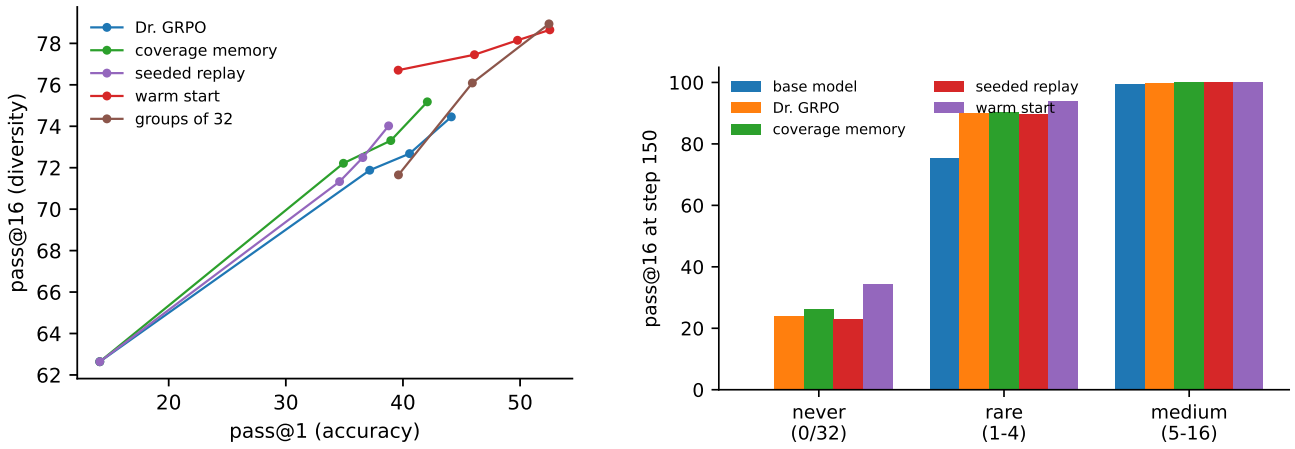


Figure 2: Left: seed-mean (pass@1, pass@16) trajectories on Qwen; S32 and G32 end near (52.5, 79), outside Dr. GRPO's 150-step range. Right: pass@16 at step 150 by base-count stratum; S32's increase over Dr. GRPO is concentrated in the $c = 0$ stratum ($24.0 \to 34.2$).

Answer budget. At 512 tokens, 48% of Dr. GRPO's sampled answers at step 150 are truncated. Re-evaluated at 1,024 tokens (Check 16, partial), Dr. GRPO seed 10 at step 150 has pass@1 57.7 and pass@16 88.4 (15.5% truncated), against 44.6 and 75.2 at 512; S32 seed 9 has 63.8 and 89.7 (4.9% truncated). Absolute values depend strongly on the budget; all comparisons are within one budget.

4 A model of the frontier (Pre-registration 17)

Model

Per-problem log-odds initialized at the empirical-Bayes estimate $\theta_x(0) = \log \frac{c_x+a}{n-c_x+b}$ (Beta(a, b) fitted by moments to the base counts) and evolving as $\dot{\theta}_x = p_x(1-p_x) + \tau N^{-1} \sum_{x'} p_{x'}(1-p_{x'})$, $p_x = \sigma(\theta_x)$: state-dependent own-problem growth plus a uniform transfer term. A checkpoint is placed at the time t where $N^{-1} \sum_x p_x(t)$ equals its observed pass@1; the model then predicts P_k for $k > 1$, stratum-level P_{16} and per-problem count likelihoods. $\tau = -0.20$ was fitted once by least squares on Dr. GRPO's nine checkpoints and frozen before the test sets were evaluated.

Error metric. RMSE $= 100\sqrt{J^{-1} \sum_j (\widehat{P}_{16,j} - P_{16,j}^{\text{model}})^2}$ pp over the J checkpoints of a test set. Registered criterion: at most 2 pp supported, 2–3 partial, above 3 fails.

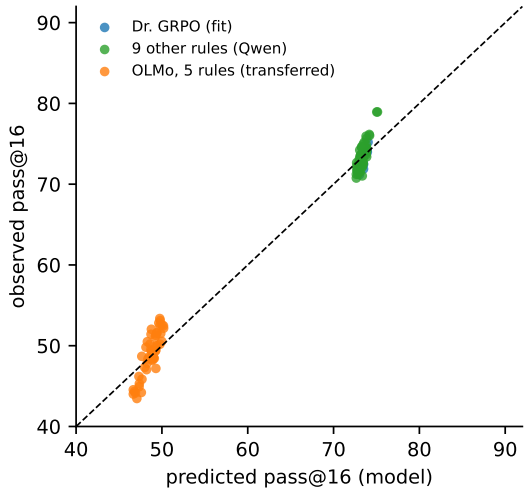


Figure 3: Model-predicted against observed pass@16. Blue: Dr. GRPO (fit). Green: nine further rules on Qwen. Orange: OLMo, five rules, with τ transferred from Qwen.

Test set (model frozen)	ckpts	RMSE: model / unif. / affine	criterion	likelihood vs affine
Fit: Dr. GRPO (in-sample)	9	1.06 / 21.29 / 1.28	–	9/9
T1: nine further rules, Qwen	56	**1.21** / 20.07 / 2.45	met	54/56
T4: OLMo, τ from Qwen	50	**1.90** / 9.85 / 12.58	met	50/50
T5: 1,024-token budget	4	**0.85** / 11.48 / 1.27	met	4/4
T2: S32, from its own step-0 counts	9	3.56 / 9.04 / 7.99	not met (> 3)	9/9
T3: 450-step Dr. GRPO (extrapolation)	–	pending	–	–

"Uniform": one log-odds shift for all problems. "Affine": $\theta' = \alpha + \beta\theta$, β fitted on the same data. "Likelihood vs affine": checkpoints where the model assigns the observed per-problem counts a higher binomial likelihood.

Reading. Within these settings, the aggregate pass@16 at matched pass@1 of policies trained from the base model is reproduced by the base count distribution and state-dependent per-problem log-odds dynamics with weak transfer; stratum residuals are small on Qwen (≤ 1.3 pp) but larger on OLMo (+6.03 never, −6.79 rare). Under a uniform log-odds shift the $c = 0$ stratum would reach 70–80 pass@16; the observed value is 24. In this model the frontier depends on two quantities, the initial count distribution and the transfer coefficient. After the warm start, the $c = 0$ stratum's mean observed-minus-predicted pass@16 is −11.92 pp, and the refitted τ is −0.65 to −0.70 in all seeds: the supervised stage changes the subsequent dynamics, not only the initial distribution.

Residuals against the model's base-initialized trajectory (an extrapolated reference beyond Dr. GRPO's range): S32 +3.3 to +3.6 pp at every checkpoint, G32 +3.9 pp at step 150 (two seeds), Dr. GRPO itself −1.2 to +0.5 pp. Whether S32 and G32 lie above Dr. GRPO's frontier depends on the extrapolation error, which T3 measures.

5 Transfer probe (Check 17, partial)

For a problem x and a fixed base-model answer y, $\ell(y) = |y|^{-1} \log \pi(y \mid x)$ (nats per token); the margin $m_x = \overline{\ell}_{\text{correct}} - \overline{\ell}_{\text{incorrect}}$ over up to four correct and four incorrect base answers with a parseable final answer, on 395 held-out problems (149 rare, 246 medium); the probe is Δm_x relative to the base model. Shared style effects need not cancel exactly. Dr. GRPO seed 8: Δm rises by +0.025 by step 150 in both strata. S32 seed 9: +0.039 in the rare stratum right after the supervised stage and +0.049 at step 150. Interim reading: Dr. GRPO's margin gain on rare problems equals its gain on medium ones in this seed, which does not support the premise that training transfers little to rare problems; four of six lanes are pending.

6 Relation to the 2025–2026 literature

- **Not found in the surveyed papers:** the noise-injection test; matched-pass@1 evaluation of advantage rules on two families; the temperature control; a frozen-parameter model predicting the matched-pass@1 frontier from base counts. None of the surveyed method papers reports matched-pass@1 evaluation (PBA, arXiv 2607.20543, includes one early-stopped control).
- **Prior papers and preprints:** per-problem transforms of p (Davis & Recht, 2510.13651) and weight-dependent convergence (Heckel et al., 2602.11128); base-seeded replay (TAR, 2610.24236); a Bayesian pass@16 gate close to our coverage weight (BBG, 2606.15455); off-policy rescue of zero-success groups (LSPO, 2607.27787; LUFFY, 2504.14945; ExGRPO, 2510.02245); transfer-driven curricula (TAC, 2606.25178; GradAlign, 2602.21492); per-prompt GRPO recurrences (Mroueh, 2503.06639); the Matthew effect in RLVR (NGU, 2609.13443); and "RLVR is a Kernel" (2609.22547), which argues that base success does not determine post-RL success and is the main challenge to the frontier model.

7 Open questions and next steps

What decides them

1. **Extrapolation of the frontier model.** T3: 450-step Dr. GRPO (seed 13 on AWS; 14 and 15 on Tamia), checkpoints beyond step 150. Deep base sampling (256 more samples on the 309 problems with $c \leq 4$) estimates the sign and uncertainty of τ.
2. **Are S32 and G32 above the frontier or further along it?** Decided by T3's extrapolation error and by Check 16 at 1,024 tokens.
3. **Method.** The pilot's group completion and prefix-guided rollouts are under evaluation. The frontier model suggests allocating demonstration acquisition to problems it predicts will stall, to be compared with random allocation and S32 at equal compute (novelty check pending).
4. **2026 methods at matched pass@1.** TAR (three seeds trained, under evaluation), BBG, MaxRL and W-REINFORCE (partially trained and evaluated).

Paper outline if T3 meets its criterion: diagnosis (per-problem equivalence of advantage rules, noise sets progress rate, matched-pass@1 protocol) → frontier model (prediction from base counts; why advantage reweighting does not move the frontier; the warm-start exception) → matched-pass@1 evaluation of 2026 methods → a method derived from the model.

A Experiment ledger

	Question	Outcome
P1–3	Down-weighting incorrect answers; collection rules at equal cost; in-training noise	noise formula fits; the optimal collection rule progresses faster along the curve
P4	PSR vs NSR, on-policy	NSR 6.4× the noise, slower, equal pass@64
P5	NSR vs PSR at matched pass@1	no gain (−2.1, −0.2 pp)
P6	The NSR paper's implementation	effect not reproduced; gradient clipping active on 26–28% of NSR steps
P7	Equal mean, graded noise (constant baselines)	progress rate ordered by predicted noise (rank correlation 0.9–0.97)
P8	pass@4 objective; reasoning graded	+1.6, +0.2 pp; reasoning-checked pass@32 equal
P9	OLMo replication	NSR−PSR −0.30 pp [−1.28, +0.77]
P10	Noise injection	PSR + trace-matched noise reproduces NSR's early lag
P11	Temperature	−1.42 pp [−2.19, −0.68]
P12	Coverage-weighted replay	+1.19 pp [+0.54, +1.79]
P14	Base-seeded replay and controls	+1.02 pp; step-150 pass@1 −5.35
P15	2026 methods at matched pass@1	running
P16	Interventions inside RL (pilot)	interim: AR and MXGC not eligible
P17	Frontier model	criterion met on T1, T4, T5; not met on T2; T3 pending
Check 16	512 vs 1,024 tokens	partial
Check 17	Transfer probe	partial

P1–P11 outcomes are from those pre-registrations' result sections (summarized in the archived story, October 20); later outcomes are in `RESULTS_12.md`, `RESULTS_17_interim.md` and `RESULTS_INTERIM_2026-10-23.md`. Registration timing, exploratory fits and data reuse are disclosed in each pre-registration.